

Amazon — first-generation embeddings-based search

2018–2021 · Semantic retrieval at the scale of a billion products

Industry collaboration · Product Search · NLP

KEYWORD MATCH

BM25 / inverted index

waterproof outdoor portable speaker

1

Waterproof Outdoor Speaker Pro X3
exact keyword match
★★★★☆ · \$78.99

2

Outdoor Speaker Waterproof Bluetooth
keyword overlap
★★★★☆ · \$65.00

3

Portable Waterproof Speaker Case
partial match — wrong product!
★★★★☆ · \$22.00

⚠ Returns cases, not speakers (keyword collision)

VS

EMBEDDING SEARCH

Dense retrieval · semantic matching

waterproof outdoor portable speaker

1

JBL Flip 6 — Portable Waterproof Speaker
semantic match · top product in category
★★★★★ · \$119.95

2

Bose SoundLink Flex · Outdoor Speaker
semantic: outdoor, durable, portable
★★★★☆ · \$149.00

3

Sony SRS-XB23 · Extra Bass · IP67
semantic: waterproof, rugged
★★★★☆ · \$89.99

✓ Returns relevant products — no false keyword matches

IMPACT

500M+ queries/day · 1B+ products · better recall for long-tail and semantic queries
Embeddings move keyword collision from common to rare — the long tail becomes reachable

Scaling semantic retrieval to industry demands is the engineering problem that makes the science real.